Ahmed Hannan

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Professional Summary

Software Engineer and Computer Science graduate (FAST-NUCES) with hands-on experience engineering scalable production web platforms and concurrent mobile applications (iOS/Android). Proficient in modern full-stack development (React, Next.js, Node.js, Django, Flutter), parallel computing (OpenMP, MPI, CUDA), and cloud databases (PostgreSQL, MongoDB, BigQuery). Awarded Performance Certificate at EmpowerBits for high-impact production engineering under concurrent user traffic. Seeking Master's studies in Computer Science to advance research in scalable distributed systems and high-performance computing architectures.

Education

National University of Computer & Emerging Sciences (FAST-NUCES), BS in Computer Science — Graduation: June 2026

- Relevant Coursework: Distributed Systems, Advanced Operating Systems, High-Performance Computing, Database Systems, Software Engineering, Networks, Algorithms.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, Dart, SQL, Bash

Frontend & Mobile: React.js, Next.js, Flutter (iOS & Android), Tailwind CSS, Redux, HTML5/CSS3, Vite

Backend & APIs: Node.js, Express.js, Django, FastAPI, Flask, RESTful APIs, Microservices Architecture

Databases & Cloud: PostgreSQL, MongoDB, Firebase, BigQuery, Redis, Google Cloud Functions

Systems & Parallel Computing: Parallel Computing (OpenMP, MPI, CUDA), Multithreading, Concurrency, Socket Programming, Linux

DevOps & Tools: Git/GitHub, Docker, CI/CD, Jira, Agile/Scrum, Selenium, Apache Airflow

Work Experience

EmpowerBits, Full-Stack Developer — 2026/06 – Present

- Engineered and deployed production-ready web and mobile applications (iOS & Android) experiencing active concurrent user loads.
- Built reactive, modular UI architectures with Flutter and React, optimizing state management and reducing re-renders to ensure smooth client performance.
- Developed resilient backend services and REST APIs, implementing database indexing, connection pooling, and caching strategies.
- Awarded Certificate of Excellence / Performance Certificate in recognition of exceptional technical contributions, engineering ownership, and product velocity.

RedCoast, Full-Stack Software Engineer — 2025/06 – 2026/06

- Spearheaded full-stack feature implementation and performance optimization across multi-tier client web and mobile applications.
- Managed one-to-one client interactions to gather operational specifications and translate them into scalable architectural designs.
- Automated integration and regression testing using Selenium and Jira, significantly improving engineering velocity and release stability.
- Conducted large-scale query optimization and data analysis on user engagement logs via Google BigQuery.

National Center of AI (NCAI, NUST), AI Intern — 2024/08 – 2024/09

- Investigated deep learning architectures with a focus on computational efficiency, resource profiling, and technical documentation.

FAST – NUCES, Teaching Assistant (Programming Fundamentals) — 2024/06 – 2024/09

- Assisted in delivering core programming coursework; mentored 50+ students in memory management, dynamic allocation, and C/C++ debugging.

Selected Projects

Pose2Play — Real-Time VR & Computer Vision Platform (Final Year Project) — GitHub: Pose2Play

- Engineered a full-stack interactive rehabilitation platform integrating low-latency computer vision pipelines with Unity VR.
- Developed real-time inter-process communication between Python kinematic estimators and Unity C# VR rendering components.

CRAG: Full-Stack Advanced RAG Platform & Evaluation Engine — GitHub: crag-advanced-rag-pipelines

- Developed a full-stack web platform implementing 7 advanced retrieval strategies (RAG Fusion, HyDE, CRAG, Graph RAG, RRR) on the Meta CRAG corpus.
- Built a Flask REST API backend integrated with FAISS vector indexing and an interactive React + Vite frontend dashboard.

Parallel Implementation of ParButterfly on Large Bipartite Graphs — HPC Graph Analytics

- Designed and implemented a scalable parallel algorithm for butterfly-motif counting in billion-edge bipartite graphs using OpenMP and MPI.
- Optimized cache locality, memory bandwidth, and load balancing across multi-core symmetric multiprocessing (SMP) and cluster environments.

GoMatch & Full-Stack Application Platforms — GitHub: GoMatch, QuizVerse

- Architected a mobile ride-hailing app replicating Uber/Careem workflows with driver-rider matching, live geolocation tracking, and trip histories.
- Engineered responsive full-stack platforms featuring role-based access control, Google OAuth/JWT authentication, and real-time state management.